

NIVELL 1:

Descàrrega els arxius CSV, estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules de les quals puguis realitzar les següents consultes:

- Crear database `transactions\_s4`.
- Marcar USE `transactions\_s4` (o doble clic al menú esquerre).
- Crear taula de fets `transaction`.

The screenshot shows a SQL IDE window with a toolbar at the top. The main editor contains the following SQL code:

```

1  -- SPRINT 4
2  -- Crear database `transactions_s4`
3  • CREATE DATABASE IF NOT EXISTS `transactions_s4`;
4
5  -- Crear taula de fets 'transaction'
6  • USE `transactions_s4`;
7  • CREATE TABLE IF NOT EXISTS transaction (
8      id VARCHAR(255),
9      card_id VARCHAR(255),
10     business_id VARCHAR(255),
11     timestamp TIMESTAMP,
12     amount decimal(10,2),
13     declined tinyint DEFAULT 0,
14     product_ids VARCHAR(255),
15     user_id VARCHAR(255),
16     lat FLOAT,
17     longitude FLOAT,
18     PRIMARY KEY (id)
19 );

```

Below the editor is an 'Output' panel with a dropdown menu set to 'Action Output'. It displays a log of executed actions:

#	Time	Action	Message
✓ 1	10:38:09	CREATE DATABASE IF NOT EXISTS `transactions_s4`	1 row(s) affected
✓ 2	10:38:17	USE `transactions_s4`	0 row(s) affected
✓ 3	10:38:17	CREATE TABLE IF NOT EXISTS transaction (id VARCHAR(255), card_id VARCHAR(255), business_id V...	0 row(s) affected

- ABANS DEL LOAD DATA, REVISAR:

- Busco si està deshabilitada (OFF) el upload desde local "SHOW GLOBAL VARIABLES LIKE 'local_infile';"
- En cas que sigui OFF, executar: "SET GLOBAL local_infile = 1;" Això activa el upload des de local temporalment fins que es reiniciï servidor
- Busco ruta on guardar arxius a carregar "SHOW VARIABLES LIKE 'secure_file_priv' ". És C:\ProgramData\MySQL\MySQL Server 8.0\Uploads\)

The screenshot shows the MySQL Workbench interface with the following SQL commands executed:

```

22 • SHOW GLOBAL VARIABLES LIKE 'local_infile'; -- Està deshabilitat (OFF) el upload desde local, llavors:
23 • SET GLOBAL local_infile = 1;
24 • SHOW VARIABLES LIKE 'secure_file_priv';

```

The **Result Grid** shows the output of the first command:

Variable_name	Value
local_infile	ON

The **Action Output** table shows the execution details:

#	Time	Action	Message
1	10:51:17	SHOW GLOBAL VARIABLES LIKE 'local_infile'	1 row(s) returned
2	10:51:17	SET GLOBAL local_infile = 1	0 row(s) affected
3	10:51:17	SHOW VARIABLES LIKE 'secure_file_priv'	1 row(s) returned

The screenshot shows the MySQL Workbench interface with the following SQL commands executed:

```

22 • SHOW GLOBAL VARIABLES LIKE 'local_infile'; -- Està deshabilitat (OFF) el upload desde local, llavors:
23 • SET GLOBAL local_infile = 1;
24 • SHOW VARIABLES LIKE 'secure_file_priv';

```

The **Result Grid** shows the output of the third command:

Variable_name	Value
secure_file_priv	C:\ProgramData\MySQL\MySQL Server 8.0\Upl...

The **Action Output** table shows the execution details:

#	Time	Action	Message
1	10:51:17	SHOW GLOBAL VARIABLES LIKE 'local_infile'	1 row(s) returned
2	10:51:17	SET GLOBAL local_infile = 1	0 row(s) affected
3	10:51:17	SHOW VARIABLES LIKE 'secure_file_priv'	1 row(s) returned

- Carregar registres des de CSV a taula 'transaction' amb LOAD. En aquest CSV, els valors estan separats per “;”. La resta de CSV d’aquesta base de dades es carregaran amb “,” (coma). Ho especificarem a FIELDS TERMINATED BY. Cada registre està acabat amb salt (“\n”) = “new line” i com tenen capçalera (nom columnes) indicarem IGNORE 1 ROWS.
- Crear taula de dimensions 'companies'.

The screenshot shows the MySQL Workbench interface. The SQL editor contains two queries. The first query is a LOAD DATA INFILE statement to load data from 'transactions.csv' into the 'transaction' table. The second query is a CREATE TABLE IF NOT EXISTS statement for the 'companies' table. The output window shows the execution results for both queries.

```

25
26 -- Carregar registres des de CSV a taula 'transaction' amb LOAD (IMPORTANT semicolon_separated_values ";")
27 • LOAD DATA
28     INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\transactions.csv'
29     INTO TABLE `transaction` -- si no fas servir "USE `DB`" abans, seria nomDB.nomTaula
30     FIELDS TERMINATED BY ';'
31     LINES TERMINATED BY '\\n'
32     IGNORE 1 ROWS;
33
34
35 -- Crear taula de dimensions 'companies'
36 • CREATE TABLE IF NOT EXISTS companies (
37     company_id VARCHAR(255),
38     company_name VARCHAR(100),
39     phone VARCHAR(20),
40     email VARCHAR(255),
41     country VARCHAR(100),
42     website VARCHAR(255),
43     PRIMARY KEY (company_id)
44 );

```

#	Time	Action	Message
1	10:51:17	SHOW GLOBAL VARIABLES LIKE 'local_infile'	1 row(s) returned
2	10:51:17	SET GLOBAL local_infile = 1	0 row(s) affected
3	10:51:17	SHOW VARIABLES LIKE 'secure_file_priv'	1 row(s) returned
4	10:53:03	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\transactions.csv' INTO TABLE...	100000 row(s) affected Records: 100000 Deleted: 0 Skipped: 0 Warnings: 0
5	10:53:11	CREATE TABLE IF NOT EXISTS companies (company_id VARCHAR(255), company_name VARCHAR(100)...	0 row(s) affected

- Carregar registres taula companies des de CSV.

The screenshot shows the MySQL Workbench interface. The SQL editor contains a single query to load data from 'companies.csv' into the 'companies' table. The output window shows the execution results for this query.

```

45
46 -- Carregar registres companies des de CSV
47 • LOAD DATA
48     INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\companies.csv'
49     INTO TABLE companies
50     FIELDS TERMINATED BY ','
51     LINES TERMINATED BY '\\n'
52     IGNORE 1 ROWS;
53

```

#	Time	Action	Message
1	10:51:17	SHOW GLOBAL VARIABLES LIKE 'local_infile'	1 row(s) returned
2	10:51:17	SET GLOBAL local_infile = 1	0 row(s) affected
3	10:51:17	SHOW VARIABLES LIKE 'secure_file_priv'	1 row(s) returned
4	10:53:03	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\transactions.csv' INTO TABLE...	100000 row(s) affected Records: 100000 Deleted: 0 Skipped: 0 Warnings: 0
5	10:53:11	CREATE TABLE IF NOT EXISTS companies (company_id VARCHAR(255), company_name VARCHAR(100)...	0 row(s) affected
6	11:17:24	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\companies.csv' INTO TABLE...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0

- Crear taula de dimensions 'credit_cards'.
- Carregar registres taula 'credit_cards' des de CSV. Com que la data del CSV ve en format diferent a l'estàndard, creem una variable d'usuari temporal per a fer la conversió de la dada a yyyy-mm-dd)

53

```

54 -- Crear taula de dimensions 'credit_cards'
55 • CREATE TABLE IF NOT EXISTS credit_cards (
56     id VARCHAR(255),
57     user_id VARCHAR(255),
58     iban VARCHAR(255),
59     pan VARCHAR(100),
60     pin VARCHAR(10),
61     cvv VARCHAR(4),
62     track1 VARCHAR(255),
63     track2 VARCHAR(255),
64     expiring_date DATE,
65     PRIMARY KEY (id)
66 );
67
68 -- Carregar registres credit_cards des de CSV
69 • LOAD DATA
70     INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\credit_cards.csv'
71     INTO TABLE credit_cards
72     FIELDS TERMINATED BY ','
73     LINES TERMINATED BY '\\n'
74     IGNORE 1 ROWS
75     (id,user_id,iban,pan,pin,cvv,track1,track2,@expiring_date)
76     SET expiring_date = str_to_date(@expiring_date, '%m/%d/%y'); -- variable d'usuari temporal q converteix dada

```

Output

#	Time	Action	Message
1	10:51:17	SHOW GLOBAL VARIABLES LIKE 'local_infile'	1 row(s) returned
2	10:51:17	SET GLOBAL local_infile = 1	0 row(s) affected
3	10:51:17	SHOW VARIABLES LIKE 'secure_file_priv'	1 row(s) returned
4	10:53:03	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\transactions.csv' INTO TABLE...	100000 row(s) affected Records: 100000 Deleted: 0 Skipped: 0 Warnings: 0
5	10:53:11	CREATE TABLE IF NOT EXISTS companies (company_id VARCHAR(255), company_name VARCHAR(100)...	0 row(s) affected
6	11:17:24	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\companies.csv' INTO TAB...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0
7	11:20:03	CREATE TABLE IF NOT EXISTS credit_cards (id VARCHAR(255), user_id VARCHAR(255), iban VARCHAR...	0 row(s) affected
8	11:20:03	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\credit_cards.csv' INTO TAB...	5000 row(s) affected Records: 5000 Deleted: 0 Skipped: 0 Warnings: 0

- Crear taula de dimensions 'users' amb una nova columna (continent) per a no perdre info del CSV.

```

78 -- Crear taula de dimensions 'users'
79 • CREATE TABLE IF NOT EXISTS users (
80     id VARCHAR(255),
81     name VARCHAR(255),
82     surname VARCHAR(255),
83     phone VARCHAR(255),
84     email VARCHAR(255),
85     birth_date DATE,
86     continent VARCHAR(255), -- NEW
87     country VARCHAR(255),
88     city VARCHAR(255),
89     postal_code VARCHAR(255),
90     address VARCHAR(255),
91     PRIMARY KEY (id)
92 );
93
94 -- Carregar registres users des de CSV
95 • LOAD DATA
96     INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\american_users.csv'
97     INTO TABLE users
98     FIELDS TERMINATED BY ','
99     ENCLOSED BY '"'
100    LINES TERMINATED BY '\\n'
101    IGNORE 1 ROWS
102    (id,name,surname,phone,email,@birth_date,country,city,postal_code,address)
103    SET
104        birth_date = str_to_date(@birth_date, '%b %d, %Y'),
105        continent = 'AMERICA';
106

```

Output

#	Time	Action	Message
7	11:20:03	CREATE TABLE IF NOT EXISTS credit_cards (id VARCHAR(255), user_id VARCHAR(255), iban VARC...	0 row(s) affected
8	11:20:03	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\credit_cards.csv' INTO T...	5000 row(s) affected Records: 5000 Deleted: 0 Skipped: 0 Warnings: 0
9	11:31:39	CREATE TABLE IF NOT EXISTS users (id VARCHAR(255), name VARCHAR(255), surname VARCHAR(...	0 row(s) affected
10	11:31:39	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\american_users.csv' INT...	1010 row(s) affected Records: 1010 Deleted: 0 Skipped: 0 Warnings: 0

The screenshot shows a MySQL Workbench window with a SQL script in the editor and its execution output in the 'Output' pane.

SQL Script:

```

107 • LOAD DATA
108   INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\european_users.csv'
109   INTO TABLE users
110   FIELDS TERMINATED BY ','
111   ENCLOSED BY '"'
112   LINES TERMINATED BY '\n'
113   IGNORE 1 ROWS
114   (id,name,surname,phone,email,@birth_date,country,city,postal_code,address)
115   SET
116     birth_date = str_to_date(@birth_date, '%b %d, %Y'),
117     continent = 'EUROPA';
118

```

Output:

#	Time	Action	Message
1	10:51:17	SHOW GLOBAL VARIABLES LIKE 'local_infile'	1 row(s) returned
2	10:51:17	SET GLOBAL local_infile = 1	0 row(s) affected
3	10:51:17	SHOW VARIABLES LIKE 'secure_file_priv'	1 row(s) returned
4	10:53:03	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\transactions.csv' INTO TABLE...	100000 row(s) affected Records: 100000 Deleted: 0 Skipped: 0 Warnings: 0
5	10:53:11	CREATE TABLE IF NOT EXISTS companies (company_id VARCHAR(255), company_name VARCHAR(100)...	0 row(s) affected
6	11:17:24	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\companies.csv' INTO TABL...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0
7	11:20:03	CREATE TABLE IF NOT EXISTS credit_cards (id VARCHAR(255), user_id VARCHAR(255), iban VARCH...	0 row(s) affected
8	11:20:03	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\credit_cards.csv' INTO TAB...	5000 row(s) affected Records: 5000 Deleted: 0 Skipped: 0 Warnings: 0
9	11:31:39	CREATE TABLE IF NOT EXISTS users (id VARCHAR(255), name VARCHAR(255), surname VARCHAR(2...	0 row(s) affected
10	11:31:39	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\american_users.csv' INTO ...	1010 row(s) affected Records: 1010 Deleted: 0 Skipped: 0 Warnings: 0
11	11:47:17	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\european_users.csv' INTO ...	3990 row(s) affected Records: 3990 Deleted: 0 Skipped: 0 Warnings: 0

- Carregar els registres a taula 'users' des de 2 CSV (American users i European users). La info de continent no ve en forma de columna a dins el CSV, però sabem que tots els registres del CSV carregat són o continent = AMERICA o continent = EUROPA. Hi ha diverses formes d'omplir la informació, però una de les que té menys passos i he fet servir és el **SET continent = X** del codi. Alternativa: al CREATE TABLE posar: "**continent VARCHAR(255) DEFAULT AMERICA**" i carregar el CSV, i en cas de NULL (tots) posaria AMERICA per defecte. Després faríem un ALTER TABLE users CHANGE/MODIFY continent VARCHAR(255) DEFAULT EUROPA, però seria uns passos més (i desavantatge que al final has d'eliminar el DEFAULT per a futurs registres).
 - o DATA: tenim un problema amb el format de la data del CSV, que ve en format "NOV 17, 1985". Per passar-ho a l'estàndard, he utilitzat el SET i la funció "string to date" que ho transforma directament durant la càrrega.

- Crear taula de dimensions 'products'

- Carregar registres 'products' des de CSV. Nota columna price: els registres venen en format \$119.22. Per a poder fer càlculs numèrics i que el dòlar no influeixi, he fet un REPLACE de \$ per " (no res) a través d'un comando SET. L'alternativa seria netejar les dades prèviament.

NOTA: Al nivell 3 crearem la taula pont i les relacions entre products / transactions / transactions_products

```

120 -- Crear taula de dimensions 'products'
121 • CREATE TABLE IF NOT EXISTS products (
122     id VARCHAR(255),
123     product_name VARCHAR(255),
124     price DECIMAL (10,2),
125     colour VARCHAR(255),
126     weight DECIMAL (10,2),
127     warehouse_id VARCHAR(255),
128     PRIMARY KEY (id)
129 );
130
131 -- Carregar registres 'products' des de CSV
132 • LOAD DATA
133     INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\products.csv'
134     INTO TABLE products
135     FIELDS TERMINATED BY ','
136     LINES TERMINATED BY '\\n'
137     IGNORE 1 ROWS
138     (id,product_name,@price,colour,weight,warehouse_id)
139     SET price = REPLACE(@price, '$', ''); -- alternativa: netejar les dades prèviament
140
141 -- NOTA: Crearem taula pont i relacions products / transactions / transactions_products al nivell 3
142
143

```

Output

#	Time	Action	Message
5	10:53:11	CREATE TABLE IF NOT EXISTS companies (company_id VARCHAR(255), company_name VARCHAR(10...	0 row(s) affected
6	11:17:24	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\companies.csv' INTO TA...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0
7	11:20:03	CREATE TABLE IF NOT EXISTS credit_cards (id VARCHAR(255), user_id VARCHAR(255), iban VARC...	0 row(s) affected
8	11:20:03	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\credit_cards.csv' INTO T...	5000 row(s) affected Records: 5000 Deleted: 0 Skipped: 0 Warnings: 0
9	11:31:39	CREATE TABLE IF NOT EXISTS users (id VARCHAR(255), name VARCHAR(255), surname VARCHAR(...	0 row(s) affected
10	11:31:39	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\american_users.csv' INT...	1010 row(s) affected Records: 1010 Deleted: 0 Skipped: 0 Warnings: 0
11	11:47:17	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\european_users.csv' INT...	3990 row(s) affected Records: 3990 Deleted: 0 Skipped: 0 Warnings: 0
12	11:51:19	CREATE TABLE IF NOT EXISTS products (id VARCHAR(255), product_name VARCHAR(255), price D...	0 row(s) affected
13	11:51:19	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\products.csv' INTO TABL...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0

- Creem les relacions de FOREIGN KEY entre transactions (taula de fets) i les taules de dimensions.

```

142
143
144 -- CREAM CONRAINTS FOREIGN KEY
145 • ALTER TABLE transaction
146     ADD FOREIGN KEY (card_id) REFERENCES credit_cards(id),
147     ADD FOREIGN KEY (business_id) REFERENCES companies(company_id),
148     ADD FOREIGN KEY (user_id) REFERENCES users(id);
149
150

```

Output

#	Time	Action	Message
1	11:57:44	ALTER TABLE transaction ADD FOREIGN KEY (card_id) REFERENCES credit_cards(id), ADD FOREIGN KE...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0

- Comprovo resultat de la càrrega dels CSV a les taules.

TAULA TRANSACTION

The screenshot shows a database interface with a SQL editor at the top containing five queries:

150 • SELECT * FROM transaction;

151 • SELECT * FROM companies;

152 • SELECT * FROM credit_cards;

153 • SELECT * FROM products;

154 • SELECT * FROM users;

155

Below the editor is the 'Result Grid' showing the results of the first query. The table has columns: id, card_id, business_id, timestamp, amount, declined, product_ids, user_id, lat, longitude. The first row is highlighted with a blue arrow.

id	card_id	business_id	timestamp	amount	declined	product_ids	user_id	lat	longitude
00043A49-2949-494B-A5DD-A5BAE38B19DD	Cc5-9294	b-2458	2024-08-28 07:16:46	395.43	0	16, 26, 97, 87	4713	46.1999	1.43554
000447FE-B650-4DCF-85DE-C7ED0EE1CAAAD	Cc5-5019	b-2370	2016-12-21 20:07:18	155.63	0	66, 69, 87	438	41.5972	12.2218
00045D6B-ED2E-4F2F-8186-CEE074D875D0	Cc5-6699	b-2390	2020-07-14 15:37:45	326.01	0	30, 11, 16, 81	2118	29.7573	-95.3796
000481C3-1C26-4FEF-83A0-4CD0EB0046BD	Cc5-6696	b-2230	2017-09-04 19:44:53	161.60	0	72	2115	53.5489	-113.503
00051AA4-9CBE-4268-B070-C38062A1B3E2	Cc5-7606	b-2266	2017-01-05 18:19:25	148.91	0	18	3025	52.2084	5.69081
0008A312-EDFE-4A4F-BC99-E9C92EC3CA4D	CcU-3358	b-2598	2023-09-23 04:51:43	294.59	0	35, 33, 19	215	53.5535	-113.499
0009A151-9RCF-4F31-9053-A468FF77FAAAR	Cc5-7509	b-7546	2023-12-31 00:06:36	383.63	0	93, 55, 28, 91	2928	51.9362	5.34265

Below the result grid is the 'Output' section, which shows the execution of the queries. The first query returned 100,000 rows, and the others returned 100, 5,000, 100, and 5,000 rows respectively.

#	Time	Action	Message
1	12:02:11	SELECT * FROM transaction	100000 row(s) returned
2	12:02:12	SELECT * FROM companies	100 row(s) returned
3	12:02:12	SELECT * FROM credit_cards	5000 row(s) returned
4	12:02:12	SELECT * FROM products	100 row(s) returned
5	12:02:12	SELECT * FROM users	5000 row(s) returned

TAULA COMPANIES

The screenshot shows a database interface with the same five SQL queries as the previous one. The 'Result Grid' shows the results of the second query, which is highlighted with a blue arrow.

company_id	company_name	phone	email	country	website
b-2222	Ac Fermentum Incorporated	06 85 56 52 33	donec.porttitor.tellus@yahoo.net	Germany	https://instagram.com/site
b-2226	Magna A Neque Industries	04 14 44 64 62	risus.donec.nibh@icloud.org	Australia	https://whatsapp.com/group/9
b-2230	Fusce Corp.	08 14 97 58 85	risus@protonmail.edu	United States	https://pinterest.com/sub/cars
b-2234	Convallis In Incorporated	06 66 57 29 50	mauris.ut@aol.couk	Germany	https://cnn.com/user/110
b-2238	Ante Iaculis Nec Foundation	08 23 04 99 53	sed.dictum.proin@outlook.ca	New Zealand	https://netflix.com/settings
b-2242	Donec Ltd	01 25 51 37 37	at.iaculis@hotmail.couk	Norway	https://nytimes.com/user/110
b-2246	Sed Nunc Ltd	02 62 64 73 48	nibh@yahoo.orn	United Kingdom	https://cnn.com/one

The 'Output' section shows the execution of the queries. The second query returned 100 rows, and the others returned 100, 5,000, 100, and 5,000 rows respectively.

#	Time	Action	Message
1	12:02:11	SELECT * FROM transaction	100000 row(s) returned
2	12:02:12	SELECT * FROM companies	100 row(s) returned
3	12:02:12	SELECT * FROM credit_cards	5000 row(s) returned
4	12:02:12	SELECT * FROM products	100 row(s) returned
5	12:02:12	SELECT * FROM users	5000 row(s) returned

TAULA CREDIT CARDS

150 • SELECT * FROM transaction;
 151 • SELECT * FROM companies;
 152 • SELECT * FROM credit_cards;
 153 • SELECT * FROM products;
 154 • SELECT * FROM users;
 155

id	user_id	iban	pan	pin	cvv	track1	track2	expiring_date
Cc5-4857	276	XX4857591835292505850771	2314242385113924	1819	467	%B2314242385113924~%WCBUDLWCBUD~22...	%B2314242385113924~2410101518363164?	2025-09-27
Cc5-4858	277	XX8581768137002436094025	6582720299715533	3964	817	%B6582720299715533~%TQMWITQMWI~2404...	%B6582720299715533~2411011049546272?	2028-12-28
Cc5-4859	278	XX7826930491423553609370	8861684536289642	4983	277	%B8861684536289642~%COFBCOCCOFBGD~280...	%B8861684536289642~2502101761665371?	2026-11-26
Cc5-4860	279	XX555990368835304645299	2481155515498459	6876	661	%B2481155515498459~%TLJTJTLJTJ~31040...	%B2481155515498459~2602101514414395?	2027-07-27
Cc5-4861	280	XX2035182877195191627307	1308930301149557	5710	398	%B1308930301149557~%POBNZPOBNZ~330...	%B1308930301149557~2805101751305028?	2026-04-25
Cc5-4862	281	XX4774721462463645409758	6715617009807829	4042	174	%B6715617009807829~%LDMWTLDMWTD~33...	%B6715617009807829~2210101702370428?	2026-11-27
Cc5-4863	282	XX1476879664545046207111	3140879819451794	5969	449	%B3140879819451794~%YXXYYXXYYX~270...	%B3140879819451794~1310101159648599?	2029-12-27

transaction 11 companies 12 credit_cards 13 products 14 users 15

Output

Action Output

#	Time	Action	Message
1	12:02:11	SELECT * FROM transaction	100000 row(s) returned
2	12:02:12	SELECT * FROM companies	100 row(s) returned
3	12:02:12	SELECT * FROM credit_cards	5000 row(s) returned
4	12:02:12	SELECT * FROM products	100 row(s) returned
5	12:02:12	SELECT * FROM users	5000 row(s) returned

TAULA PRODUCTS

150 • SELECT * FROM transaction;
 151 • SELECT * FROM companies;
 152 • SELECT * FROM credit_cards;
 153 • SELECT * FROM products;
 154 • SELECT * FROM users;
 155

id	product_name	price	colour	weight	warehouse_id
1	Direwolf Stannis	161.11	#7c7c7c	1.00	WH-4
10	Karstark Dorne	119.52	#f4f4f4	2.40	WH-5
100	south duel	40.43	#6d6d6d	3.00	WH-95
11	Karstark Dorne	49.70	#141414	2.70	WH-6
12	duel Direwolf	181.60	#a8a8a8	2.10	WH-7
13	palpatine chewbacca	139.59	#2b2b2b	1.00	WH-8
14	Direwolf	147.53	#c4c4c4	2.00	WH-9

transaction 11 companies 12 credit_cards 13 products 14 users 15

Output

Action Output

#	Time	Action	Message
1	12:02:11	SELECT * FROM transaction	100000 row(s) returned
2	12:02:12	SELECT * FROM companies	100 row(s) returned
3	12:02:12	SELECT * FROM credit_cards	5000 row(s) returned
4	12:02:12	SELECT * FROM products	100 row(s) returned
5	12:02:12	SELECT * FROM users	5000 row(s) returned

TAULA USERS

150 • SELECT * FROM transaction;
 151 • SELECT * FROM companies;
 152 • SELECT * FROM credit_cards;
 153 • SELECT * FROM products;
 154 • SELECT * FROM users;
 155

id	name	surname	phone	email	birth_date	continent	country	city	postal_code	address
1	Zeus	Gamble	1-282-581-0551	interdum.enim@protonmail.edu	1985-11-17	AMERICA	United States	New York	10001	348-7818 Sagittis St.
10	Robert	Mccarthy	(324) 746-6771	fermentum.protonmail.com	1984-04-30	AMERICA	United States	San Jose	95101	P.O. Box 773
100	Melodie	Mclean	1-677-221-7152	risus.varius@google.ca	1989-09-15	AMERICA	United States	San Jose	95101	Ap #644-8492 Sagittis St.
1000	Amikrv	Qbuixbp	+48-258-9936	amikrv.qbuixbp@example.com	1970-05-17	EUROPA	Germany	Stuttgart	70173	215 Qbuixbp St
1001	Nfvrib	Oydaivbg	+94-121-2522	nfvrib.oydaivbg@example.com	1994-03-04	EUROPA	Germany	Cologne	50667	121 Oydaivbg St
1002	Jbdfmd	Jbdfdhvp	+70-120-3668	jbdfmd.jbdfdhvp@example.com	2001-09-27	EUROPA	Germany	Munich	80331	412 Jbdfdhvp St
1003	Juvrin	Sfrfvmzi	+58-173-6968	juvrin.sfrfvmzi@example.com	1981-01-20	EUROPA	Germany	Stuttgart	70173	735 Sfrfvmzi St

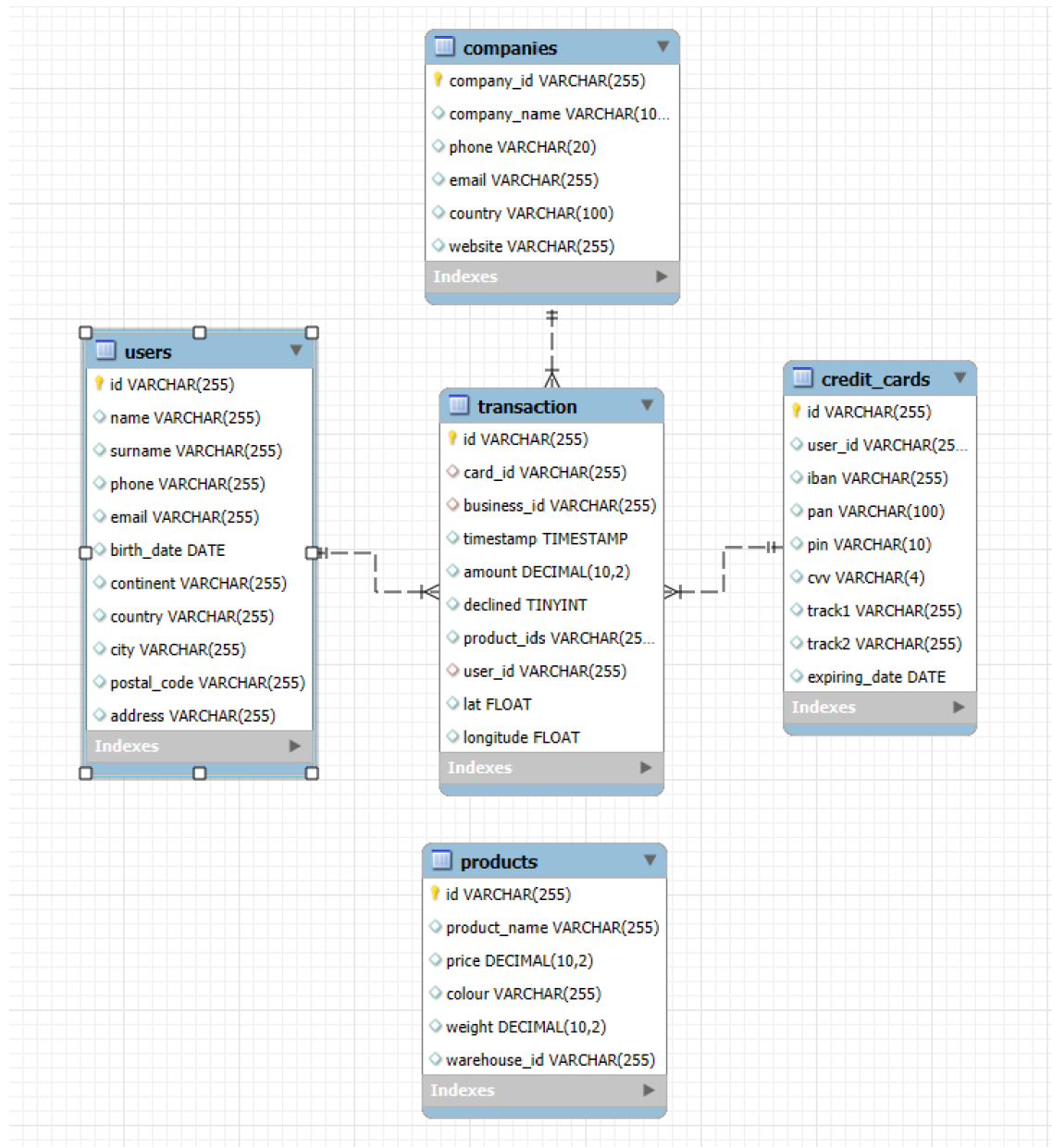
transaction 11 companies 12 credit_cards 13 products 14 users 15

Output

Action Output

#	Time	Action	Message
1	12:02:11	SELECT * FROM transaction	100000 row(s) returned
2	12:02:12	SELECT * FROM companies	100 row(s) returned
3	12:02:12	SELECT * FROM credit_cards	5000 row(s) returned
4	12:02:12	SELECT * FROM products	100 row(s) returned
5	12:02:12	SELECT * FROM users	5000 row(s) returned

- Comprovo resultat creació de la base de dades, les taules i els constraints.



EXERCICI 1. Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules.

```

156 -- Nivell 1 EX 1: subconsulta usuaris > 80 transaccions (min. 2 taules)
157 • SELECT id, name AS Nom, surname AS Cognom, email
158 FROM users
159 WHERE id IN (
160     SELECT user_id
161     FROM transaction
162     GROUP BY user_id
163     HAVING COUNT(id) > 80
164 )
165 ORDER BY id;
166

```

Result Grid

id	Nom	Cognom	email
185	Molly	Gilliam	donec@outlook.couk
289	Dxwgi	Hwcru	dxwgi.hwcru@example.com
318	Bnyr	Astuw	bnyr.astuw@example.com
454	Sfzoh	Xgvfridxs	sfzoh.xgvfridxs@example.com
NULL	NULL	NULL	NULL

users 17 x

Output

Action Output

#	Time	Action	Message
1	12:11:18	SELECT id, name AS Nom, surname AS Cognom, email FROM users WHERE id IN (SELECT user_id FROM ...	4 row(s) returned

EXERCICI 2. Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.

```

166
167
168 -- Nivell 1 EX 2: mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, (min. 2 taules)
169 • SELECT c.company_id, iban, ROUND(AVG(amount),2) AS mitjana
170 FROM transaction t
171 INNER JOIN companies c ON business_id = c.company_id
172 INNER JOIN credit_cards cc ON cc.id = card_id
173 WHERE company_name LIKE ('%Donec Ltd%')
174 GROUP BY iban, c.company_id
175 ORDER BY mitjana DESC; -- per veure major gasto 1er
176

```

Result Grid

company_id	iban	mitjana
b-2242	XX383017813919620199366352	680.69
b-2242	XX637706357397570394973913	680.01
b-2242	XX971393971465292202312259	645.46
b-2242	XX171847116928892375969307	628.89
b-2242	XX225424638818542406223575	608.68
b-2242	XX748890729057195711766071	607.29
b-2242	TN9614563570667381893122	605.41
b-2242	XX481908034037364242591185	605.36
b-2242	XX194675519739256335753508	597.19

Result 34 x

Output

Action Output

#	Time	Action	Message
1	17:05:03	SELECT c.company_id, iban, ROUND(AVG(amount),2) AS mitjana FROM transaction t INNER JOIN companies ...	371 row(s) returned

NIVELL 2:

Crea una nova taula que reflecteixi l'estat de les targetes de crèdit basat en: si les tres últimes transaccions han estat declinades aleshores és inactiu, si almenys una no és rebutjada aleshores és actiu. Partint d'aquesta taula respon: **EXERCICI 1:** [Quantes targetes estan actives?](#)

ALTERNATIVA 1. La més ràpida (més eficient), és fent servir una declaració WITH (CTE) + una funció de finestra (row_number) per fer un "ranking" de transaccions en ordre descendent per data (timestamp) i agrupat (partició) per cada card_id. Amb un SELECT i subqueries faig el tall a les 3 últimes transaccions de cada targeta, estableixo el CASE inactiva si les 3 últimes transaccions hagin estat declined = 1, i faig el count / recompte.

```

178 -- Nivell 2 EX 1: estat targetes
179
180 -- Crear taula
181 CREATE TABLE targetes_actives (
182   card_id VARCHAR(255),
183   Activitat_targeta VARCHAR(50)
184 );
185
186 -- Fer filtre targetes per inserir registres a la taula.
187 INSERT INTO targetes_actives (card_id, Activitat_targeta)
188
189 -- ALTERNATIVA AMB FUNCIONS DE FINESTRA (row_number) I CTE
190 WITH ranking_targetes AS (SELECT
191   card_id,
192   id AS id_transaccio,
193   timestamp AS data_transaccio,
194   declined,
195   ROW_NUMBER() OVER(PARTITION BY card_id ORDER BY timestamp DESC) AS ranking_estat_targetes
196 FROM transaccio)

```

card_id	Activitat_targeta
Cs5-4857	Targeta ACTIVA
Cs5-4858	Targeta ACTIVA
Cs5-4859	Targeta ACTIVA
Cs5-4860	Targeta ACTIVA
Cs5-4861	Targeta ACTIVA

targetes_actives 40 x Result 41

#	Time	Action	Message	Duration / Fetch
1	17:47:08	CREATE TABLE targetes_actives (card_id VARCHAR(255), Activitat_targeta VARCHAR(50))	0 row(s) affected	0.047 sec
2	17:47:08	INSERT INTO targetes_actives (card_id, Activitat_targeta) -- ALTERNATIVA AMB FUNCIONS DE FINESTRA...	4995 row(s) affected Records: 4995 Duplicates: 0 Warnings: 0	0.641 sec
3	17:47:08	SELECT * FROM targetes_actives	4995 row(s) returned	0.000 sec / 0.000 sec
4	17:47:08	SELECT count(Activitat_targeta) AS recompte_targetes_actives FROM targetes_actives	1 row(s) returned	0.000 sec / 0.000 sec

Continuació:

```

197 SELECT card_id, Activitat_targeta
198 FROM (
199   SELECT
200     card_id,
201     CASE
202       WHEN SUM(declined) > 3 THEN 'Targeta INACTIVA'
203       ELSE 'Targeta ACTIVA'
204     END AS 'Activitat_targeta'
205   FROM (
206     SELECT
207       card_id,
208       id_transaccio,
209       data_transaccio,
210       declined,
211       ranking_estat_targetes
212     FROM ranking_targetes
213     WHERE ranking_estat_targetes <= 3
214     ) AS taula_ranking_3
215   GROUP BY card_id
216   ORDER BY card_id
217   ) AS classificacio_activa_inactiva
218 WHERE Activitat_targeta = 'Targeta ACTIVA';
219
220
221 SELECT * FROM targetes_actives; -- comprovar inserció registres
222 SELECT count(Activitat_targeta) AS recompte_targetes_actives
223 FROM targetes_actives; -- RESULTAT RECOMpte AMB WITH=ROW_NUMBER()

```

recompte_targetes_actives
4995

targetes_actives 40 x Result 41 x

#	Time	Action	Message	Duration / Fetch
1	17:47:08	CREATE TABLE targetes_actives (card_id VARCHAR(255), Activitat_targeta VARCHAR(50))	0 row(s) affected	0.047 sec
2	17:47:08	INSERT INTO targetes_actives (card_id, Activitat_targeta) -- ALTERNATIVA AMB FUNCIONS DE FINESTRA...	4995 row(s) affected Records: 4995 Duplicates: 0 Warnings: 0	0.641 sec
3	17:47:08	SELECT * FROM targetes_actives	4995 row(s) returned	0.000 sec / 0.000 sec
4	17:47:08	SELECT count(Activitat_targeta) AS recompte_targetes_actives FROM targetes_actives	1 row(s) returned	0.000 sec / 0.000 sec

ALTERNATIVA 2. Amb subqueries.

-La subquery 1 al WHERE obté una llista de transaccions per card_id i declined (0/1) APLICANT UN FILTRE de les 3 transaccions més recents a partir d'un recompte (si tenen 0, 1 o 2 transaccions més recents, formaran part de la llista).

-La subquery 2 al FROM obté una llista de classificació ACTIVA / INACTIVA segons el criteri a CASE del SELECT, agrupant per card_id.

-La query exterior fa la llista per INSERTAR registres a la taula.

NOTA: el criteri CASE l'he establert com a ACTIVA ≤ 2 , ja que per a que sigui considerada inactiva, les 3 últimes transaccions han de ser DECLINED = 1(true), és a dir, SUMA de 1declined + declined+ 1 declined=3

-La query final (COUNT) fa un recompte a la nova taula amb quantes targetes tenen estat = ACTIVA.

```

224
225 -- ALTERNATIVA AMB SUBQUERIES (més lenta)
226
227 DROP TABLE targetes_actives;
228
229
230 CREATE TABLE targetes_actives (
231     card_id VARCHAR(255),
232     Activitat_targeta VARCHAR(50)
233 );
234
235
236 INSERT INTO targetes_actives (card_id, Activitat_targeta)
237 SELECT *
238 FROM
239 (
240     SELECT
241         card_id,
242         CASE
243             WHEN SUM(declined) = 3 THEN 'Targeta INACTIVA'
244             ELSE 'Targeta ACTIVA'
245         END AS Activitat_targeta
246     FROM
247     (
248         SELECT
249             t1.card_id,
250             t1.declined
251         FROM transaction t1
252         WHERE
253         (
254             SELECT COUNT(*)
255             FROM transaction t2
256             WHERE t2.card_id = t1.card_id
257               AND t2.timestamp > t1.timestamp -- per cada transaccio segons card_id recompta num trans amb data més recent
258             ) < 3 -- filtra 0, 1 o 2 registres més recents (top3)
259         ) AS 3_recents
260         GROUP BY card_id
261         ORDER BY card_id asc
262     ) AS taula_targetes_actives_inactives
263     WHERE Activitat_targeta = 'Targeta ACTIVA';
264
265 SELECT * FROM targetes_actives; -- comprovar inserció registres
266 SELECT count(Activitat_targeta) AS recompte_targetes_actives FROM targetes_actives; -- RESULTAT RECOMpte AMB SUBQUERIES
267
268

```

Result Grid			
recompte_targetes_actives			
4995			
targetes_actives 42 Result 43			
Output			
Action Output			
#	Time	Action	Message
1	17:47:08	CREATE TABLE targetes_actives (card_id VARCHAR(255), Activitat_targeta VARCHAR(50))	0 row(s) affected
2	17:47:08	INSERT INTO targetes_actives (card_id, Activitat_targeta) - ALTERNATIVA AMB FUNCIONS DE FINESTRA (...)	4995 row(s) affected Records: 4995 Duplicates: 0 Warnings: 0
3	17:47:08	SELECT * FROM targetes_actives	4995 row(s) returned
4	17:47:08	SELECT count(Activitat_targeta) AS recompte_targetes_actives FROM targetes_actives	1 row(s) returned
5	17:58:49	DROP TABLE targetes_actives	0 row(s) affected
6	17:59:21	CREATE TABLE targetes_actives (card_id VARCHAR(255), Activitat_targeta VARCHAR(50))	0 row(s) affected
7	17:59:21	INSERT INTO targetes_actives (card_id, Activitat_targeta) SELECT * FROM (SELECT card_id, CASE WHEN ...	4995 row(s) affected Records: 4995 Duplicates: 0 Warnings: 0
8	17:59:30	SELECT * FROM targetes_actives	4995 row(s) returned
9	17:59:30	SELECT count(Activitat_targeta) AS recompte_targetes_actives FROM targetes_actives	1 row(s) returned

NIVELL 3:

Crea una taula amb la qual puguem unir les dades del nou arxiu products.csv amb la base de dades creada, tenint en compte que des de transaction tens product_ids. Genera la següent consulta: **EXERCICI 1:** *Necessitem conèixer el nombre de vegades que s'ha venut cada producte.*

a) Creo taula intermitja 'transaction_product' amb id_compost dels dos (no serà columna)

```

243
244 -- NIVELL 3
245
246 -- Crear taula intermitja 'transaction_product' amb id_compost dels dos (no serà columna)
247 CREATE TABLE IF NOT EXISTS transaction_product (
248     transaction_id VARCHAR(255) NOT NULL,
249     product_id VARCHAR(255) NOT NULL,
250     price DECIMAL (10,2),
251     unitats INT UNSIGNED,
252     PRIMARY KEY (transaction_id, product_id),
253     FOREIGN KEY (product_id) REFERENCES products(id),
254     FOREIGN KEY (transaction_id) REFERENCES transaction(id)
255 );
256
257 SELECT * FROM transaction_product;

```

transaction_id	product_id	price	unitats
NULL	NULL	NULL	NULL

transaction_product26 x

Output

Action Output

#	Time	Action	Message
1	12:51:00	CREATE TABLE IF NOT EXISTS transaction_product (transaction_id VARCHAR(255) NOT NULL, produc...	0 row(s) affected
2	12:51:00	SELECT * FROM transaction_product	0 row(s) returned

b) incloure els registres a la taula pont (2 alternatives): **FIND_IN_SET()** i **JSON_TABLE()**

OPCIÓ 1: FIND_IN_SET(). La funció find_in_set() fa unió i posa el registre a la llista d'insertar si products_ids a transactions coincideixen amb products.id. El replace transforma a llista sense espais per buscar sense problemes.

CONTRES: lent en cas de moltes dades i no funcionaria si una transacció tingués varies unitats del mateix producte.

259 -- Carrego els registres productes de la transacció a la taula pont.
 260 -- ALTERNATIVA 1
 261 * INSERT INTO transaction_product (transaction_id, product_id, price, unitats)
 262 SELECT
 263 t.id AS transaction_id,
 264 p.id AS product_id,
 265 p.price AS price,
 266 1 AS unitats -- si posem DEFAULT 1 al CREATE TABLE, ja no caldria
 267 FROM transaction t
 268 JOIN products p
 269 ON FIND_IN_SET(p.id, REPLACE(t.product_ids, ' ', '')) > 0
 270 WHERE t.declined = 0;
 271
 272 * select * from transaction_product;

transaction_id	product_id	price	unitats
00043A49-2949-494B-ASDO-ASBAE3B819D0	16	180.91	1
00043A49-2949-494B-ASDO-ASBAE3B819D0	26	53.01	1
00043A49-2949-494B-ASDO-ASBAE3B819D0	87	96.26	1
00043A49-2949-494B-ASDO-ASBAE3B819D0	97	65.25	1
000447FE-6650-4DCF-8SDE-C7ED0EE1CAAD	66	2.12	1
000447FE-6650-4DCF-8SDE-C7ED0EE1CAAD	69	57.25	1
000447FE-6650-4DCF-8SDE-C7ED0EE1CAAD	87	96.26	1
00045D6B-ED2E-4F2F-8186-CEE074D875D0	11	49.70	1
00045D6B-ED2E-4F2F-8186-CEE074D875D0	16	180.91	1

Output

#	Time	Action	Message	Duration / Fetch
1	13:00:59	INSERT INTO transaction_product (transaction_id, product_id, price, unitats) SELECT 1.id AS transaction_id,...	252693 row(s) affected Records: 252693 Duplicates: 0 Warnings: 0	4.296 sec

274 -- suma total d'unitats venudes agrupada per product_id
 275 * SELECT
 276 product_id AS Ref_producto,
 277 SUM(unitats) AS total_unitats_venudes -- també ok COUNT(*) o COUNT(unitats) pq unitat = 1 per registre
 278 FROM transaction_product
 279 JOIN transaction ON transaction_id = id
 280 WHERE declined = 0
 281 GROUP BY Ref_producto
 282 ORDER BY total_unitats_venudes; -- veure 1er els top vendes

Ref_producto	total_unitats_venudes
43	2405
42	2419
22	2437
19	2446
37	2452
71	2452
69	2453
77	2453
75	2454
51	2459

Output

#	Time	Action	Message	Duration / Fetch
1	13:00:59	INSERT INTO transaction_product (transaction_id, product_id, price, unitats) SELECT 1.id AS transaction_id,...	252693 row(s) affected Records: 252693 Duplicates: 0 Warnings: 0	4.296 sec
2	13:01:03	select * from transaction_product	252693 row(s) returned	0.000 sec / 0.094 sec
3	13:02:14	SELECT product_id AS Ref_producto, SUM(unitats) AS total_unitats_venudes -- també ok COUNT(*) o COU...	100 row(s) returned	0.360 sec / 0.000 sec

Eliminem taula per fer-ho tot amb l'opció JSON_TABLE(). No es poden eliminar només registres per restriccions MYSQL.

OPCIÓ 2. JSON_TABLE()

```

283
284 DROP TABLE transaction_product;
285
286 -- ALTERNATIVA 2: PER SERVER JSON_TABLE
287 INSERT INTO transaction_product (transaction_id, product_id, price, unitats)
288 SELECT
289     t.id AS transaction_id,
290     jt.product_id,
291     p.price AS price,
292     1 AS unitats
293 FROM
294     transaction t,
295     JSON_TABLE
296     (
297         CONCAT('{"', REPLACE(REPLACE(t.product_ids, ' ', ''), ',', ', '), '"}', -- no cal si tingues format ["73","21","43"]
298         "$[*]") COLUMNS (product_id VARCHAR(255) PATH "$")
299     ) AS jt
300 JOIN products p ON p.id = jt.product_id
301 WHERE t.declined = 0;
302
303 -- el find_in_set es aprox. 1 segon més lent que JSON_TABLE

```

#	Time	Action	Message	Duration / Fetch
1	13:00:59	INSERT INTO transaction_product (transaction_id, product_id, price, unitats) SELECT t.id AS transaction_id...	252693 row(s) affected Records: 252693 Duplicates: 0 Warnings: 0	4.296 sec
2	13:01:03	select * from transaction_product	252693 row(s) returned	0.000 sec / 0.094 sec
3	13:02:14	SELECT product_id AS Ref_producto, SUM(unitats) AS total_unitats_venudes -- també ok COUNT(*) o COU...	100 row(s) returned	0.360 sec / 0.000 sec
4	13:04:30	DROP TABLE transaction_product	0 row(s) affected	0.031 sec
5	13:04:37	CREATE TABLE IF NOT EXISTS transaction_product (transaction_id VARCHAR(255) NOT NULL, produc...	0 row(s) affected	0.031 sec
6	13:04:48	INSERT INTO transaction_product (transaction_id, product_id, price, unitats) SELECT t.id AS transaction_id...	252693 row(s) affected Records: 252693 Duplicates: 0 Warnings: 0	3.453 sec

```

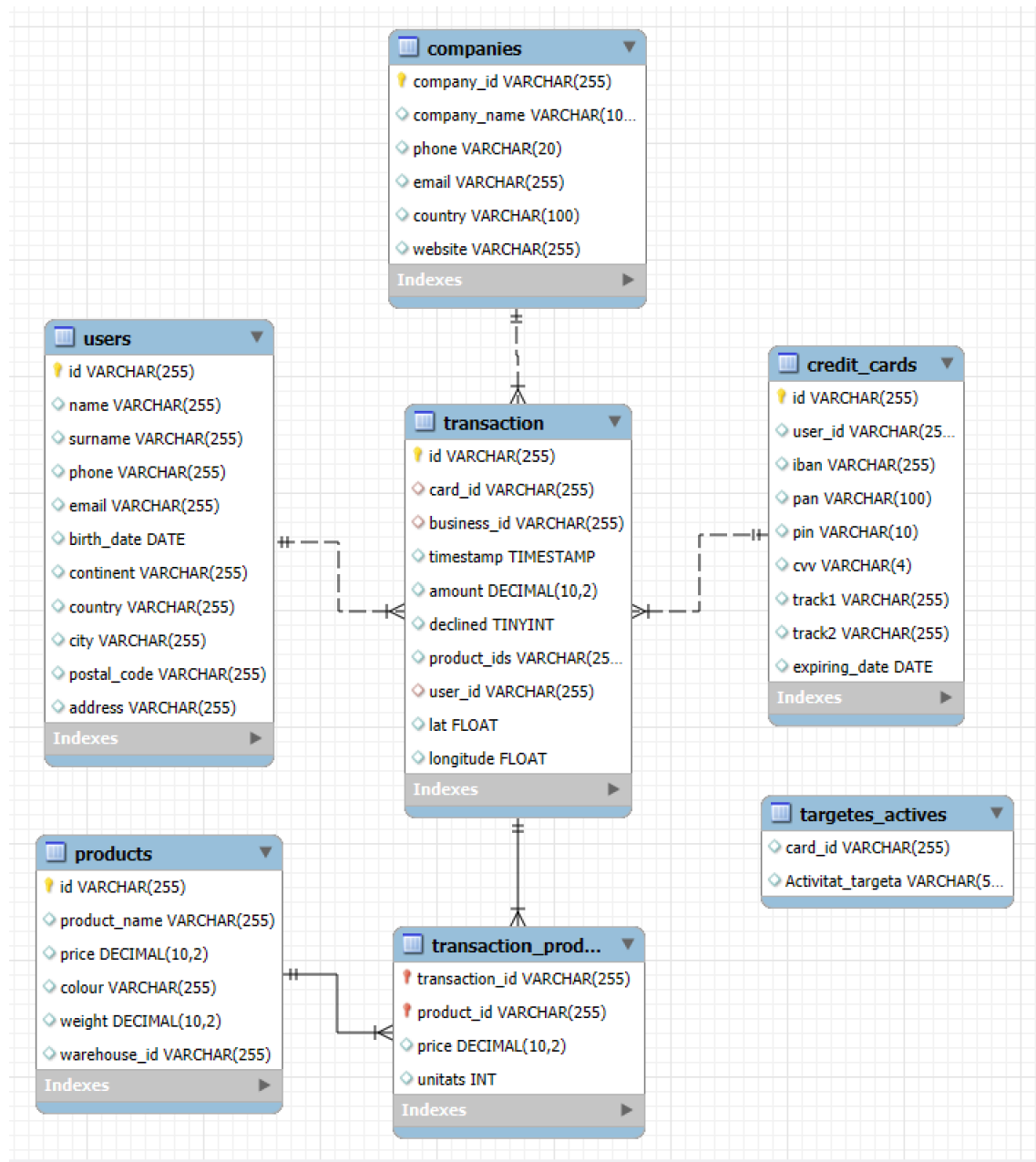
274 -- suma total d'unitats venudes agrupada per product_id
275 SELECT
276     product_id AS Ref_producto,
277     SUM(unitats) AS total_unitats_venudes -- també ok COUNT(*) o COUNT(unitats) pq unitat = 1 per registre
278 FROM transaction_product
279 JOIN transaction ON transaction_id = id
280 WHERE declined = 0
281 GROUP BY Ref_producto
282 ORDER BY total_unitats_venudes; -- veure l'er els top vende

```

Ref_producto	total_unitats_venudes
43	2405
42	2419
22	2437
19	2446
37	2452
71	2452
69	2453
77	2453
75	2454
51	2459

#	Time	Action	Message	Duration / Fetch
1	13:00:59	INSERT INTO transaction_product (transaction_id, product_id, price, unitats) SELECT t.id AS transaction_id...	252693 row(s) affected Records: 252693 Duplicates: 0 Warnings: 0	4.296 sec
2	13:01:03	select * from transaction_product	252693 row(s) returned	0.000 sec / 0.094 sec
3	13:02:14	SELECT product_id AS Ref_producto, SUM(unitats) AS total_unitats_venudes -- també ok COUNT(*) o COU...	100 row(s) returned	0.360 sec / 0.000 sec
4	13:04:30	DROP TABLE transaction_product	0 row(s) affected	0.031 sec
5	13:04:37	CREATE TABLE IF NOT EXISTS transaction_product (transaction_id VARCHAR(255) NOT NULL, produc...	0 row(s) affected	0.031 sec
6	13:04:48	INSERT INTO transaction_product (transaction_id, product_id, price, unitats) SELECT t.id AS transaction_id...	252693 row(s) affected Records: 252693 Duplicates: 0 Warnings: 0	3.453 sec
7	13:05:41	SELECT product_id AS Ref_producto, SUM(unitats) AS total_unitats_venudes -- també ok COUNT(*) o COU...	100 row(s) returned	0.359 sec / 0.000 sec

Nota: he considerat només transaccions NO declinades (declined = 0).

SCHEMA FINAL**PEER TO PEER**

NOTES: Trinidad ha fet la revisió peer to peer.

Correccions afegides:

- Sugerència nivell 1 ex 2: afegir company_id al SELECT per assegurar.
- Falta screenshot final schema incloent taula pont.